

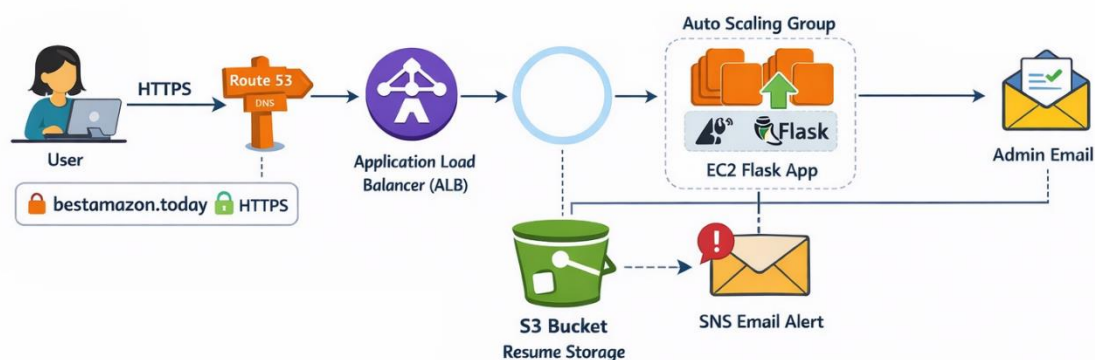
Resume Upload Application on AWS

Project Overview: This project is a Resume (CV) Upload Web Application built using AWS cloud services. The main goal of this project is to allow users to upload their resume securely and notify the admin automatically.

This project is a **web application** where:

- A user opens a website using a custom domain:
<https://bestamazon.today>
- The user enters their **name, email**, and uploads their **resume in PDF format**
- The CV is stored safely in **Amazon S3**
- Once the resume is uploaded, an **email notification is sent to the admin** using **Amazon SNS**
- The application is designed to be **highly available, scalable, and secure**

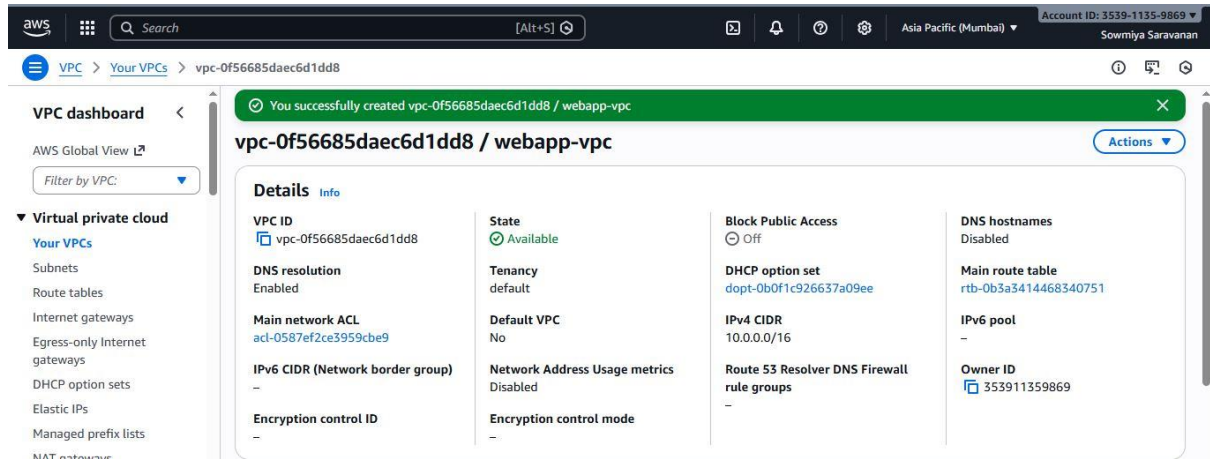
Resume Upload Application on AWS



PHASE 1 – Create VPC (Networking Base)

Step 1: Create VPC

1. Go to **VPC** → **Your VPCs** → **Create VPC**
2. Settings:
 - **Name:** webapp-vpc
 - **IPv4 CIDR:** 10.0.0.0/16
3. Click **Create VPC**



Step 2: Create Subnets

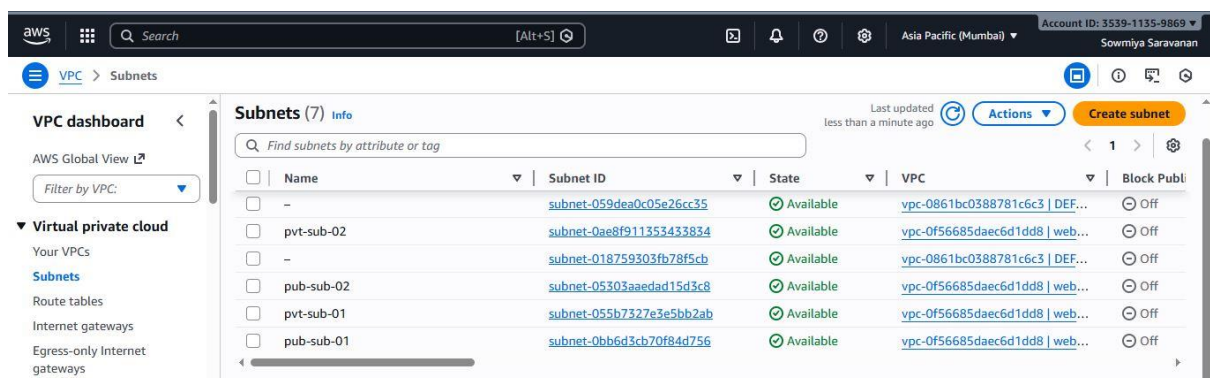
You need **2 Public + 2 Private subnets**.

Public Subnets

- pub-sub-01: **10.0.1.0/24** → AZ: ap-south-1a
- pub-sub-02: **10.0.2.0/24** → AZ: ap-south-1b

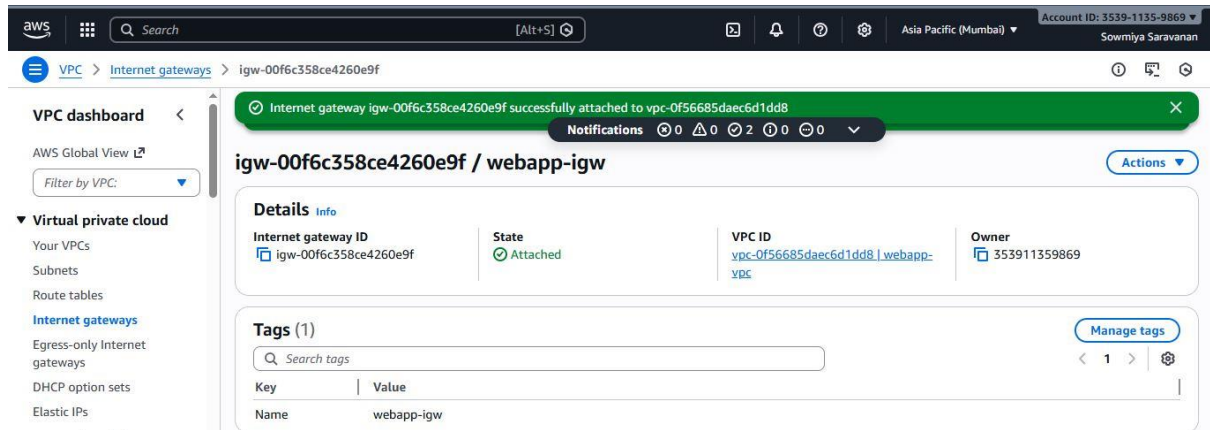
Private Subnets

- pvt-sub-01: **10.0.11.0/24** → AZ: ap-south-1a
- pvt-sub-02: **10.0.12.0/24** → AZ: ap-south-1c



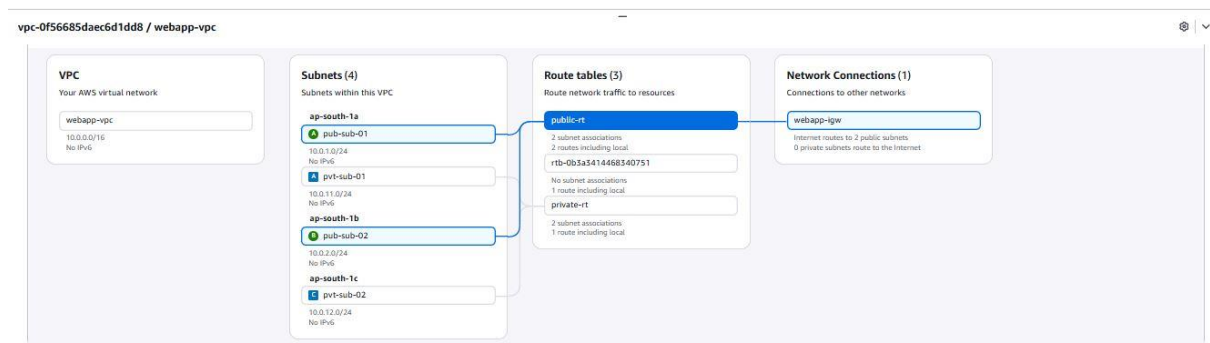
Step 3: Create Internet Gateway

1. Go to **VPC** → **Internet Gateways** → **Create**
2. Name: **webapp-igw**
3. Attach IGW → choose **webapp-vpc**



Step 4: Create Route Table (Public)

1. Create Route Table → Name: **public-rt**
2. Edit Routes → Add
 - Destination: **0.0.0.0/0**
 - Target: **Internet Gateway**
3. Associate this Route Table to both **Public Subnets**



STEP 5: Create EC2 Instance

AWS Console → EC2 → Launch instance

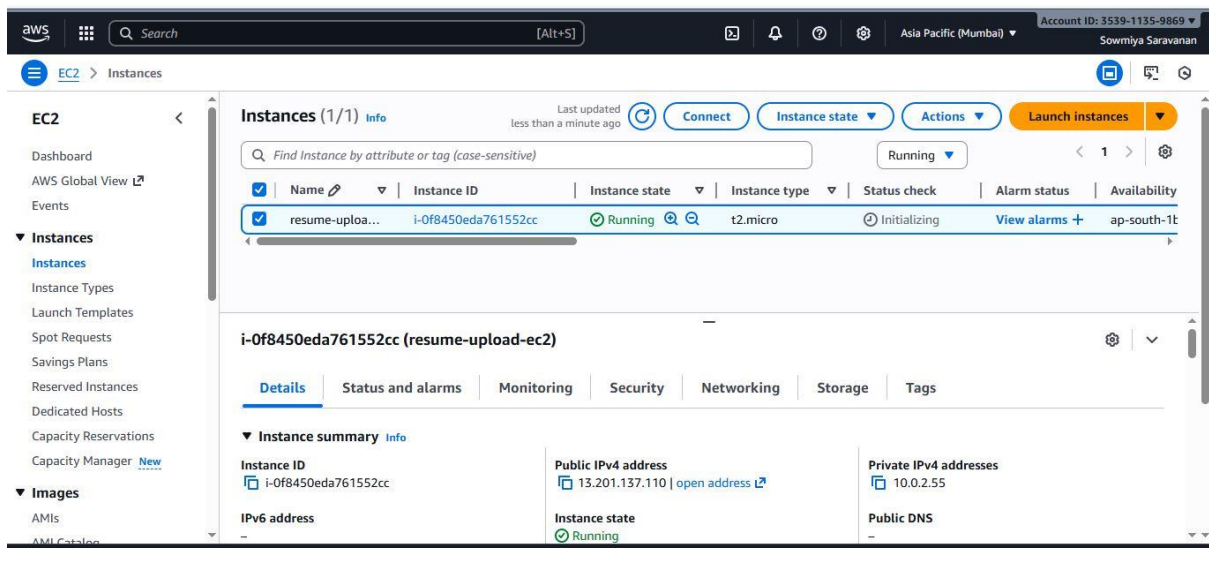
Details:

- Name: **resume-upload-ec2**
- AMI: **Amazon Linux 2023**
- Instance type: **t2.micro**
- Key pair: **Your keypair name**
- Network: **Your VPC**
- Auto-assign public IP: **ENABLE**

Security Group (Inbound):

- SSH – 22 – My IP
- HTTP – 80 – 0.0.0.0/0

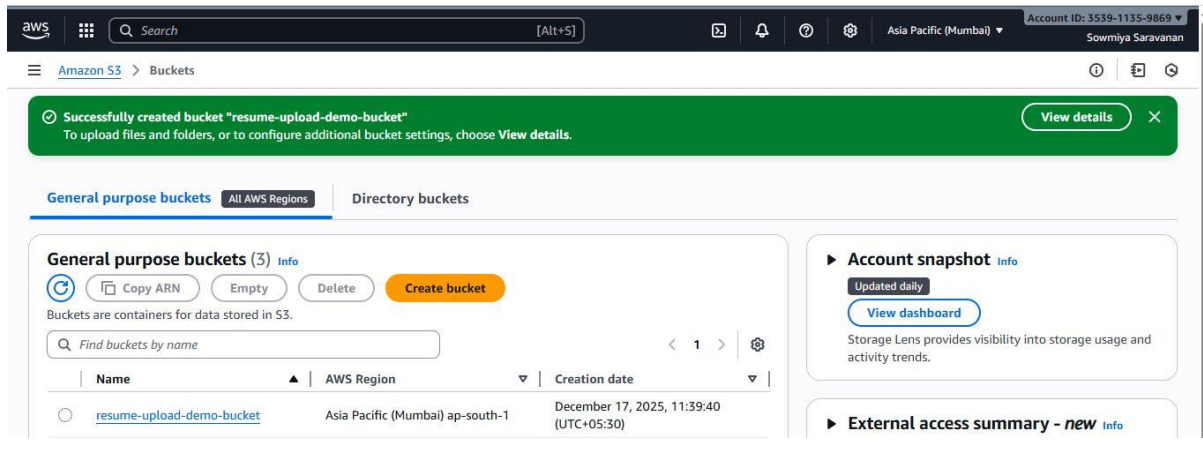
☞ Launch instance



STEP 6: Create S3 Bucket

AWS → S3 → Create bucket

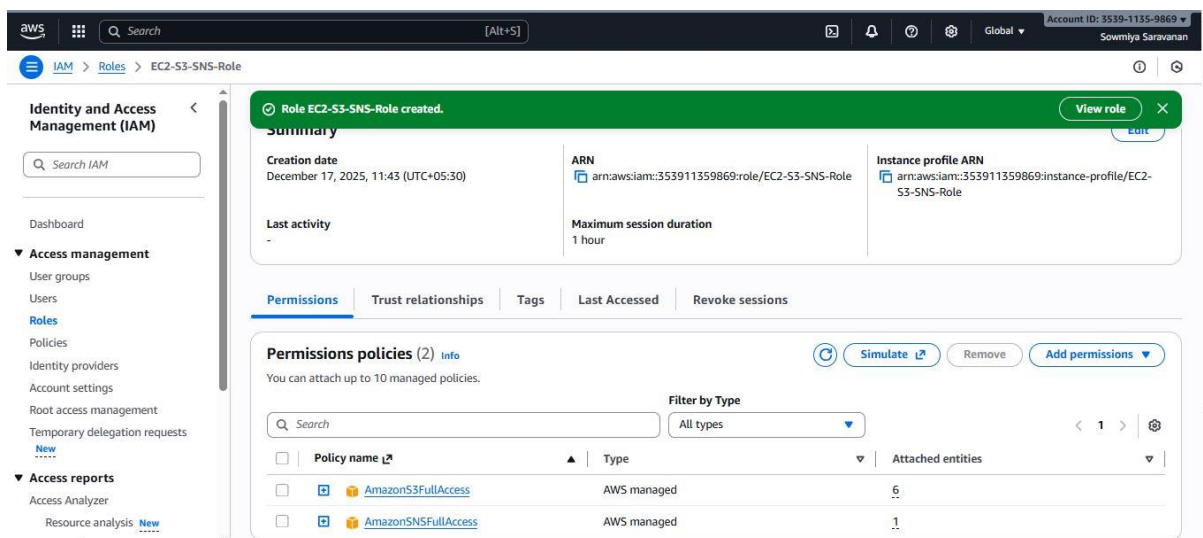
- Bucket name: **resume-upload-demo-bucket**
- Region: same as EC2
- Block public access: ON
- Create bucket



STEP 7: Create IAM Role & Attach to EC2

IAM → Roles → Create role

- Trusted entity: EC2
- Policies:
 - AmazonS3FullAccess
 - AmazonSNSFullAccess
- Role name: EC2-S3-SNS-Role



Attach role:

EC2 → Instance → Actions → Modify IAM Role → select role

AWS Search [Alt+S] Asia Pacific (Mumbai) Sowmiya Saravanan Account ID: 3539-1135-9869

EC2 > Instances

EC2

Dashboard
AWS Global View
Events

Instances (1/2) Info Last updated less than a minute ago

Successfully attached EC2-S3-SNS-Role to instance i-01773c0485a580568

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
resume-uploa...	i-01773c0485a580568	Running	t2.micro	2/2 checks pass	View alarms +	ap-south-1a

```
app = Flask(__name__)
```

```
BUCKET_NAME = "resume-upload-demo-bucket"
```

```
SNS_TOPIC_ARN = "PASTE_SNS_TOPIC_ARN_HERE"
```

```
s3 = boto3.client("s3")
```

```
sns = boto3.client("sns")
```

```
HTML = """
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Resume Upload</title>
```

```
<style>
```

```
body { background:#020617; font-family:Arial; height:100vh;
```

```
display:flex; justify-content:center; align-items:center; }
```

```
.box { background:white; padding:25px; border-radius:12px; width:360px; }
```

```
button { width:100%; padding:10px; background:#16a34a; color:white;
```

```
border:none; }
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="box">
```

```
<h2>Upload Resume</h2>
```

```
<form method="POST" enctype="multipart/form-data">
```

```
<input name="name" placeholder="Name" required><br><br>
```

```
<input name="email" type="email" placeholder="Email" required><br><br>
```

```
<input type="file" name="file" accept="application/pdf" required><br><br>
```

```
<button>Upload</button>
```

```
</form>
```

```
<p>{{msg}}</p>
```

```
</div>
```

```
</body>
```

```
</html>
```

```
"""
```

```
@app.route("/", methods=["GET", "POST"])
```

```
def upload():
```

```
    msg=""
```

```
    if request.method=="POST":
```

```
        name=request.form["name"]
```

```
        email=request.form["email"]
```



```

file=request.files["file"]

if file.filename.endswith(".pdf"):
    ts=datetime.now().strftime("%Y%m%d%H%M%S")
    fname=f"{name}_{ts}.pdf"
    path="/tmp/"+fname
    file.save(path)

    s3.upload_file(path, BUCKET_NAME, fname)
    sns.publish(
        TopicArn=SNS_TOPIC_ARN,
        Subject="New Resume Uploaded",
        Message=f"Name: {name}\nEmail: {email}\nFile: {fname}"
    )
    msg="Resume uploaded & email sent!"
else:
    msg="Only PDF allowed"

return render_template_string(HTML,msg=msg)

app.run(host="0.0.0.0",port=80)

```

Save & exit.

```

"""
@app.route("/", methods=["GET", "POST"])
def upload():
    msg=""
    if request.method=="POST":
        name=request.form["name"]
        email=request.form["email"]
        file=request.files["file"]

        if file.filename.endswith(".pdf"):
            ts=datetime.now().strftime("%Y%m%d%H%M%S")
            fname=f"{name}_{ts}.pdf"
            path="/tmp/"+fname
            file.save(path)

            s3.upload_file(path, BUCKET_NAME, fname)
            sns.publish(
                TopicArn=SNS_TOPIC_ARN,
                Subject="New Resume Uploaded",
                Message=f"Name: {name}\nEmail: {email}\nFile: {fname}"
            )
            msg="Resume uploaded & email sent!"
        else:
            msg="Only PDF allowed"

    return render_template_string(HTML,msg=msg)

app.run(host="0.0.0.0",port=80)
-- INSERT --

```

67,1

Bot

STEP 8: Test App Manually

sudo python3 app.py


```
[root@ip-10-0-2-55 resume-app]# sudo python3 app.py
/usr/local/lib/python3.9/site-packages/boto3/compat.py:89: PythonDeprecationWarning: Boto3 will no longer support Python 3
.9 starting April 29, 2026. To continue receiving service updates, bug fixes, and security updates please upgrade to Pytho
n 3.10 or later. More information can be found here: https://aws.amazon.com/blogs/developer/python-support-policy-updates-
for-aws-sdks-and-tools/
  warnings.warn(warning, PythonDeprecationWarning)
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:80
* Running on http://10.0.2.55:80
Press CTRL+C to quit
```

PHASE 3: AUTO START (systemd)

STEP 9: Create systemd Service

sudo vi /etc/systemd/system/resumeapp.service

```
[Unit]
Description=Resume Upload Flask App
After=network.target

[Service]
User=root
WorkingDirectory=/root/resume-app
ExecStart=/usr/bin/python3 /root/resume-app/app.py
Restart=always

[Install]
WantedBy=multi-user.target
```

```
sudo systemctl daemon-reload
sudo systemctl enable resumeapp
sudo systemctl start resumeapp
```

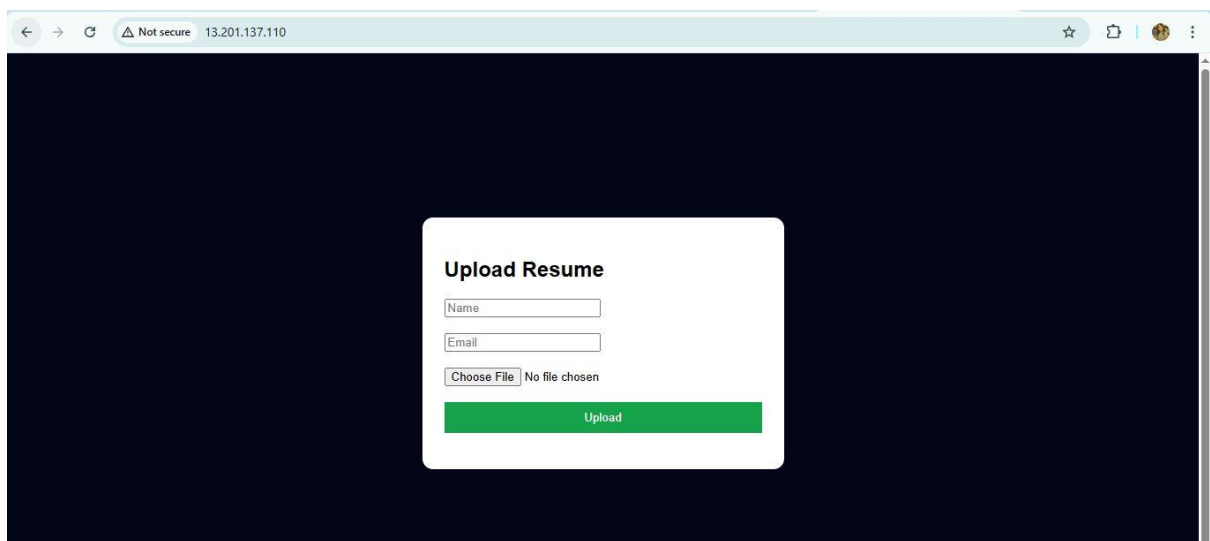
Check status:

```
sudo systemctl status resumeapp
```

```
[root@ip-10-0-2-55 resume-app]# sudo systemctl status resumeapp
● resumeapp.service - Resume Upload Flask App
   Loaded: loaded (/etc/systemd/system/resumeapp.service; enabled; preset: disabled)
   Active: active (running) since Wed 2025-12-17 06:55:41 UTC; 14s ago
     Main PID: 3096 (python3)
        Tasks: 1 (limit: 1120)
      Memory: 45.1M
         CPU: 402ms
    CGroup: /system.slice/resumeapp.service
            └─3096 /usr/bin/python3 /root/resume-app/app.py

Dec 17 06:55:41 ip-10-0-2-55.ap-south-1.compute.internal systemd[1]: Started resumeapp.service - Resume Upload Flask App.
Dec 17 06:55:41 ip-10-0-2-55.ap-south-1.compute.internal python3[3096]: /usr/local/lib/python3.9/site-packages/boto3/comp
Dec 17 06:55:41 ip-10-0-2-55.ap-south-1.compute.internal python3[3096]: warnings.warn(warning, PythonDeprecationWarning)
Dec 17 06:55:41 ip-10-0-2-55.ap-south-1.compute.internal python3[3096]: * Serving Flask app 'app'
Dec 17 06:55:41 ip-10-0-2-55.ap-south-1.compute.internal python3[3096]: * Debug mode: off
Dec 17 06:55:41 ip-10-0-2-55.ap-south-1.compute.internal python3[3096]: WARNING: This is a development server. Do not use
Dec 17 06:55:41 ip-10-0-2-55.ap-south-1.compute.internal python3[3096]: * Running on all addresses (0.0.0.0)
Dec 17 06:55:41 ip-10-0-2-55.ap-south-1.compute.internal python3[3096]: * Running on http://127.0.0.1:80
Dec 17 06:55:41 ip-10-0-2-55.ap-south-1.compute.internal python3[3096]: * Running on http://10.0.2.55:80
Dec 17 06:55:41 ip-10-0-2-55.ap-south-1.compute.internal python3[3096]: Press CTRL+C to quit
lines 1-20/20 (END)
```

Browser: <http://13.201.137.110/>

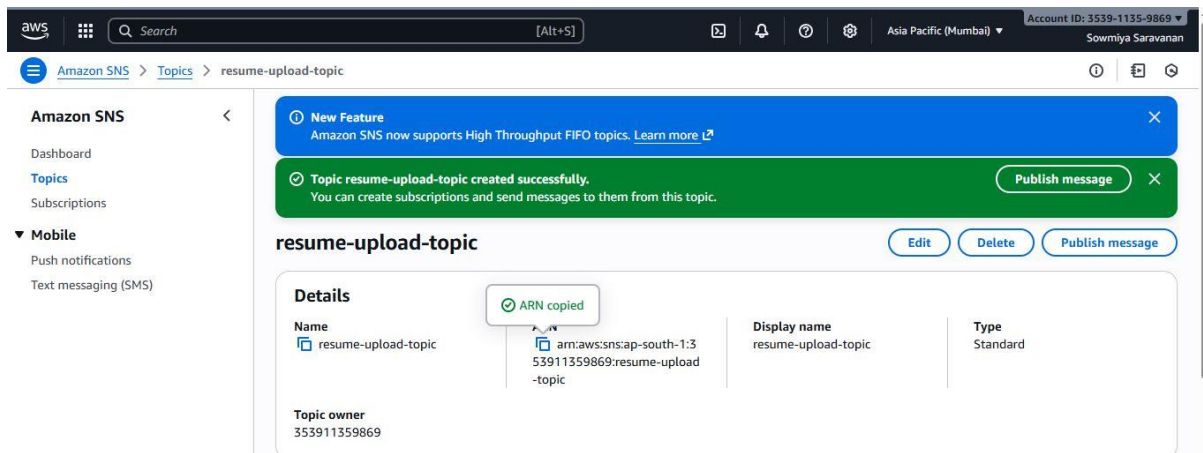


The screenshot shows a web browser window with the address bar displaying "Not secure 13.201.137.110". The main content area features a dark blue background with a white rectangular form in the center. The form is titled "Upload Resume" and contains three input fields: "Name", "Email", and a file selection button labeled "Choose File" with the text "No file chosen" next to it. Below these fields is a green button labeled "Upload".

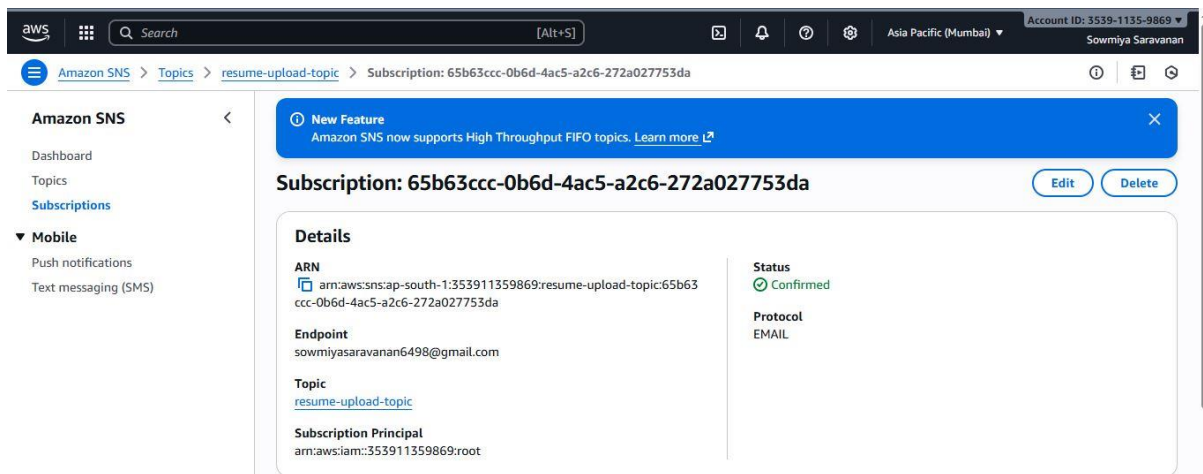
PHASE 4: SNS EMAIL SETUP

STEP 10: Create SNS Topic

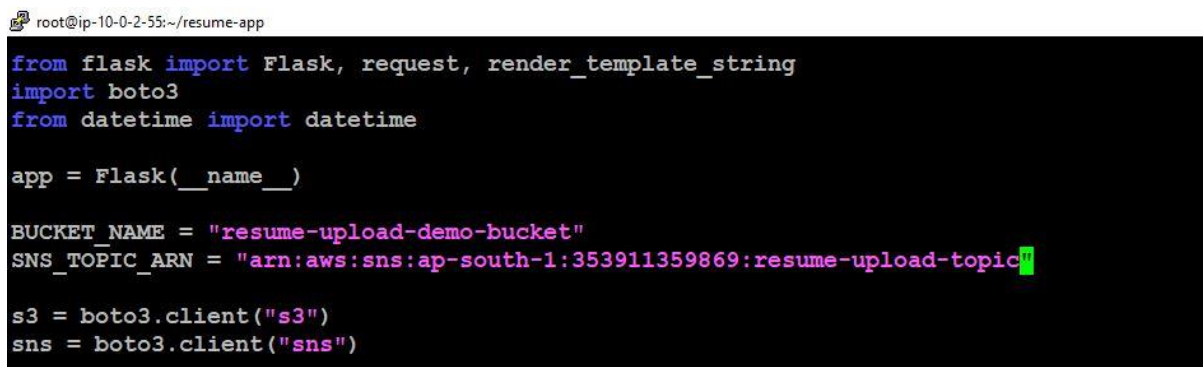
SNS → Topics → Create topic
Name: resume-upload-topic



Create email subscription → Confirm email



Copy Topic ARN → paste in app.py



Restart service:

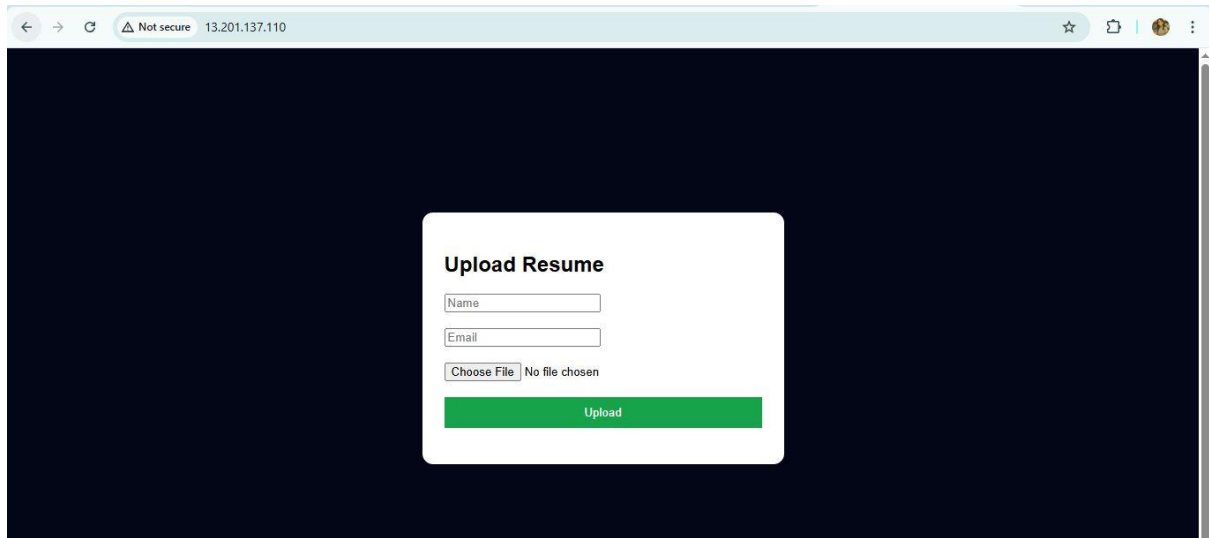
`sudo systemctl daemon-reload`

`sudo systemctl restart resumeapp`

`sudo systemctl status resumeapp`

sudo systemctl restart resumeapp

Open Browser: <http://13.201.137.110/>

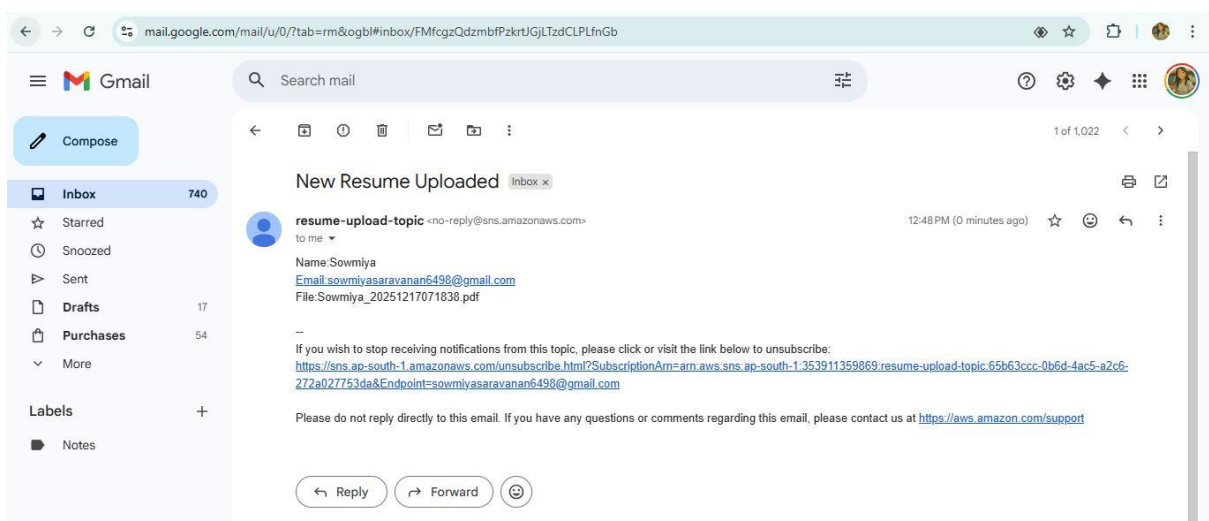


The screenshot shows a web browser window with the address bar displaying "13.201.137.110". The main content area has a dark blue background. In the center, there is a white rectangular box titled "Upload Resume". Inside this box, there are three input fields: "Name", "Email", and a file selection button labeled "Choose File" with the text "No file chosen" next to it. Below these fields is a green button labeled "Upload".

Then Upload your resume here,

- Name: Test
- Email: your mail
- File: any PDF
- Click **Upload**

Then SNS will trigger the email,



PHASE 5: ALB + AUTO SCALING

STEP 11: Create AMI

EC2 → Instance → Actions → Create image

Name: resume-upload-ami

The screenshot shows the AWS Management Console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and account information (Asia Pacific (Mumbai), Account ID: 3539-1135-9869, Sowmiya Saravanan). The main content area is titled 'EC2 > Instances'. A green notification banner at the top states: 'Currently creating AMI ami-05e67b3f8e0bfc0a4 from instance i-0f8450eda761552cc. Check that the AMI status is 'Available' before deleting the instance or carrying out other actions related to this AMI.' Below this, the 'Instances (1/2)' table shows one instance: 'resume-uploa...' with Instance ID 'i-0f8450eda761552cc', state 'Running', type 't2.micro', and '2/2 checks passed'. The left sidebar shows the navigation menu with 'Instances' selected. Below the instance list, the 'Amazon Machine Images (AMIs) (1)' section is visible, showing the newly created AMI 'resume-upload-ami' with ID 'ami-05e67b3f8e0bfc0a4'.

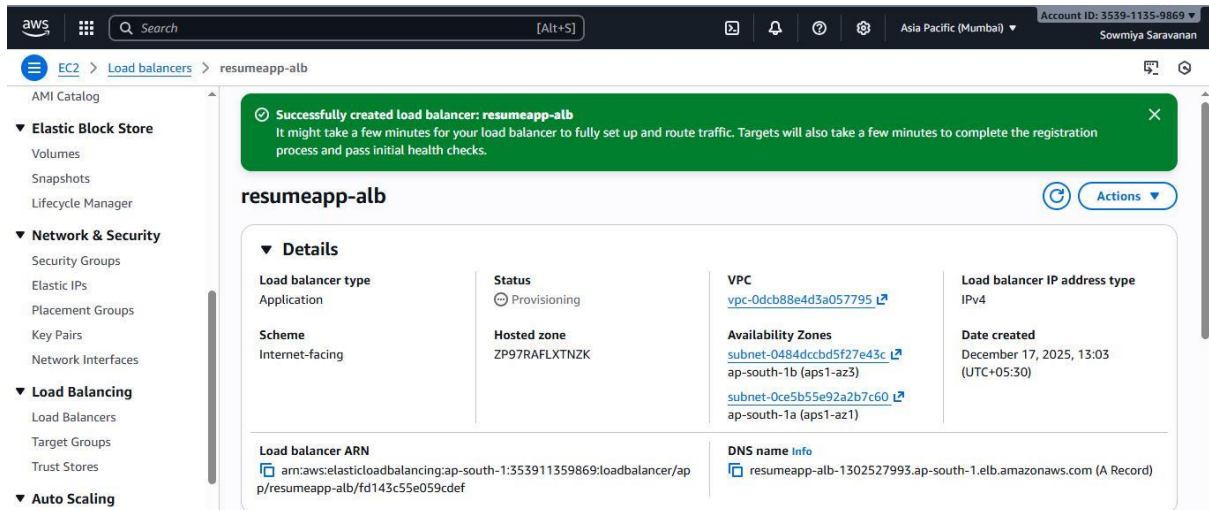
STEP 12: Create Target Group

- Type: Instance
- Port: 80
- Name: resumeapp-tg
- Health check: /

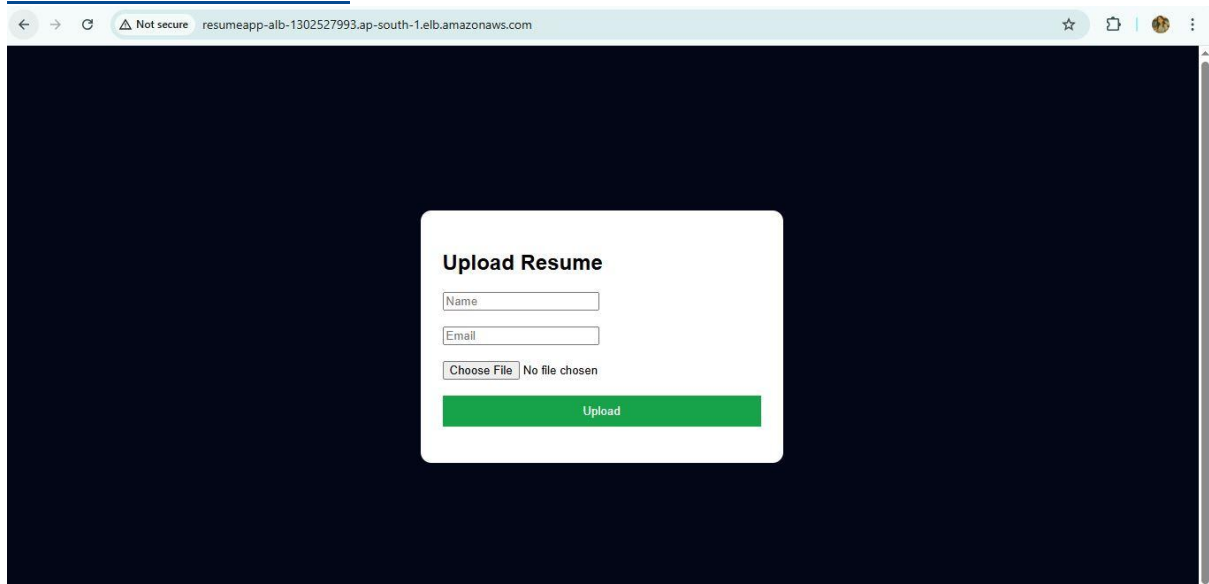
The screenshot shows the AWS Management Console interface for the 'resumeapp-tg' target group. A green notification banner at the top states: 'Successfully created the target group: resumeapp-tg. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the Targets tab.' The 'Details' section shows the target group's ARN, target type 'Instance', protocol 'HTTP', port '80', and protocol version 'HTTP1'. It also shows the VPC ID 'vpc-0dc88e4d3a057795'. Below the details, a summary row shows: '0 Total targets', '0 Healthy', '0 Unhealthy', '0 Unused', '0 Initial', and '0 Draining'. The left sidebar shows the navigation menu with 'Load Balancing' selected.

STEP 13: Create Application Load Balancer

1. Name: **resumeapp-alb**
2. Scheme: **Internet-facing**
3. Subnets: **Both Public Subnets**
4. Security Group: **alb-sg**
5. Listener: **HTTP 80 → Forward → Target Group**



Then open a browser to hit the url: <http://resumeapp-alb-1302527993.ap-south-1.elb.amazonaws.com/>



STEP 14: Create Launch Template

- AMI: resume-upload-ami

- IAM role: EC2-S3-SNS-Role
- Security group: HTTP from ALB SG

The screenshot shows the AWS Management Console interface for a Launch Template named 'resumeapp-lt' (ID: lt-0649b83a7681ef63b). The left sidebar shows the navigation menu with 'EC2' selected. The main content area displays the 'Launch template details' and 'Launch template version details'.

Launch template details:

- Launch template ID: lt-0649b83a7681ef63b
- Launch template name: resumeapp-lt
- Default version: 1
- Owner: arn:aws:iam::353911359869:root

Launch template version details (Version 1):

- Version: 1 (Default)
- Description: -
- Date created: 2025-12-17T07:36:55.000Z
- Created by: arn:aws:iam::353911359869:root

Instance details:

- AMI ID: ami-05e67b3f8e0bfc0a4
- Instance type: t2.micro
- Availability Zone: -
- Availability Zone Id: ap-s1-az3
- Key pair name: -
- Security groups: -
- Security group IDs: -

STEP 15: Create Auto Scaling Group

- Desired: 2
- Min: 2
- Max: 4
- Attach ALB + Target Group

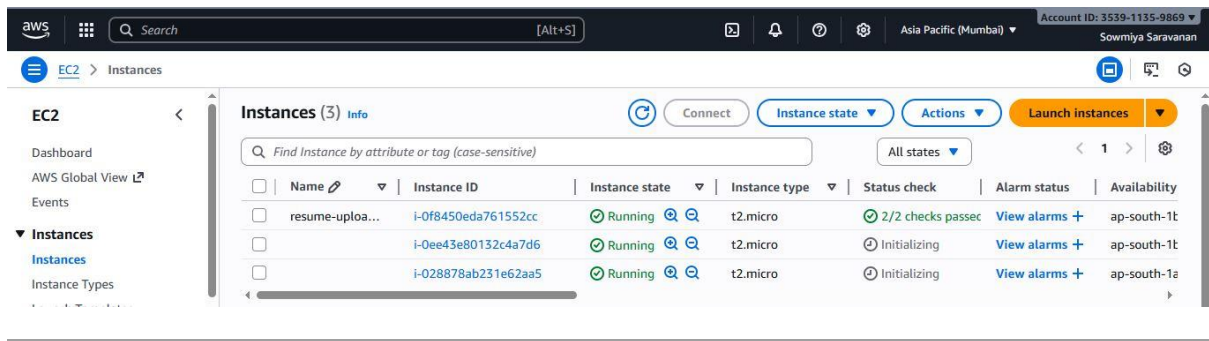
The screenshot shows the AWS Management Console interface for an Auto Scaling Group named 'resumeapp-asg'. The left sidebar shows the navigation menu with 'EC2' selected. The main content area displays the 'resumeapp-asg Capacity overview' and 'Launch template' details.

resumeapp-asg Capacity overview:

- ARN: arn:aws:autoscaling:ap-south-1:353911359869:autoScalingGroup:ec64f20c-1a22-45a7-a01b-a0df8e6b7396:autoScalingGroupName/resumeapp-asg
- Desired capacity: 2
- Scaling limits: 2 - 4
- Desired capacity type: Units (number of instances)
- Status: -
- Date created: Wed Dec 17 2025 13:14:23 GMT+0530 (India Standard Time)

Launch template:

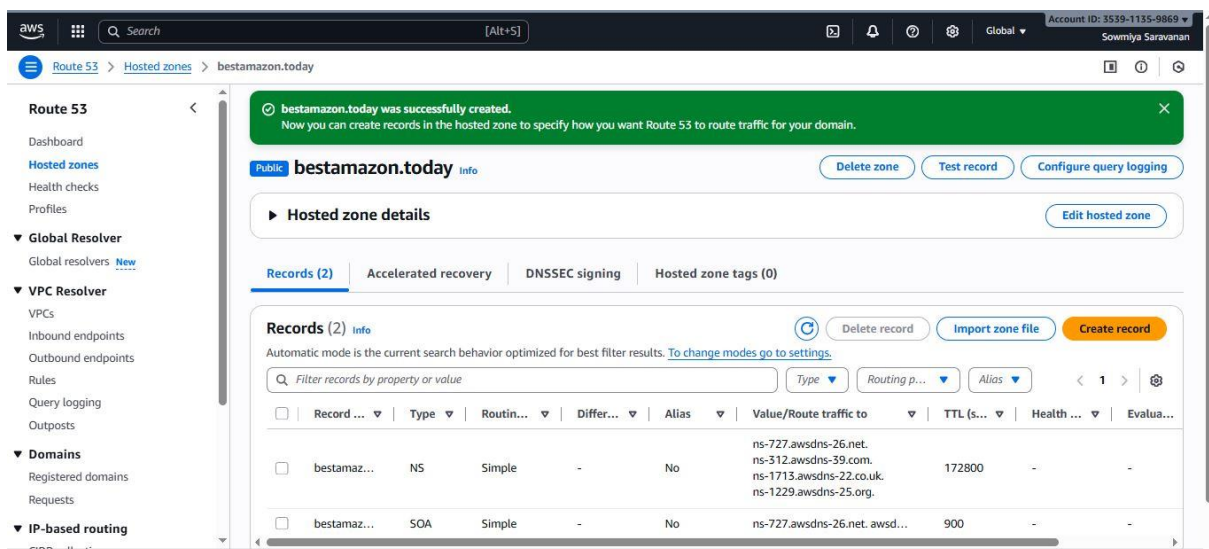
- Launch template: lt-0649b83a7681ef63b (resumeapp-lt)
- AMI ID: ami-05e67b3f8e0bfc0a4
- Instance type: t2.micro
- Owner: arn:aws:iam::353911359869:root



PHASE 6: HTTPS + DOMAIN

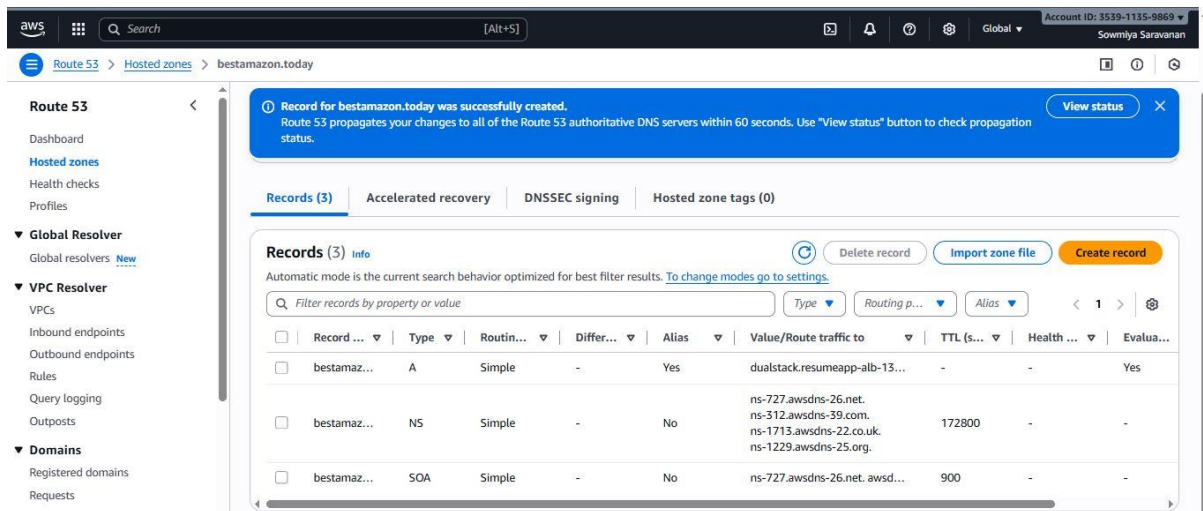
STEP 16: Route 53 Alias Record

Create a Hosted Zone for your domain name,

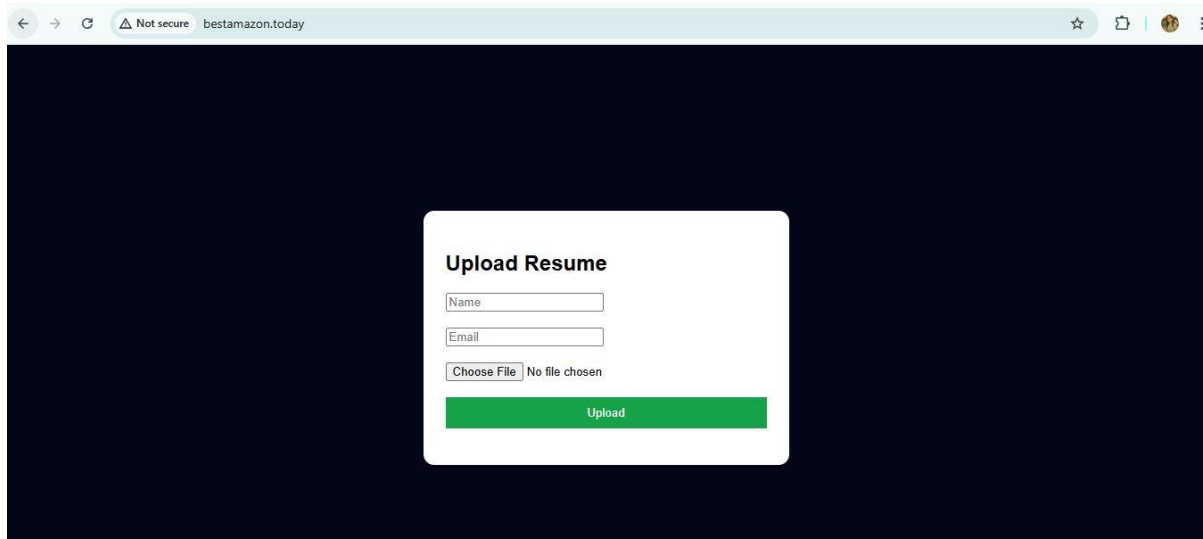


Then create a “A” record for your domain DNS.

- Alias → ALB
- Domain: bestamazon.today



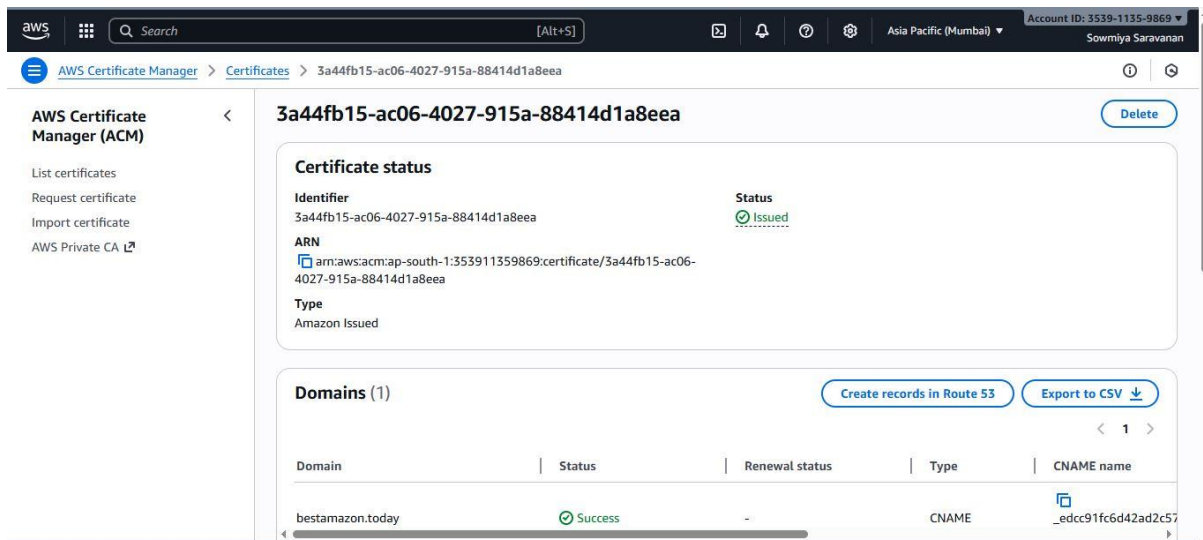
Open browser: <http://bestamazon.today>



STEP 17: Request ACM Certificate

ACM (same region as ALB)

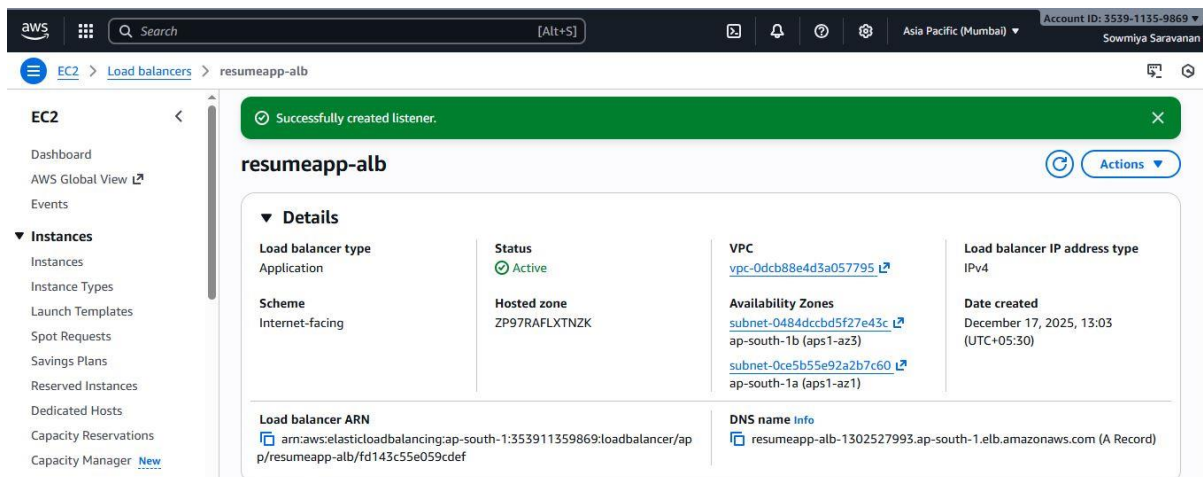
- Domain: bestamazon.today
- Validation: DNS → Create record in Route53
- Status: Issued



STEP 18: Add HTTPS Listener

ALB → Add listener

- HTTPS 443
- Attach ACM cert
- Forward to target group



Update ALB SG:

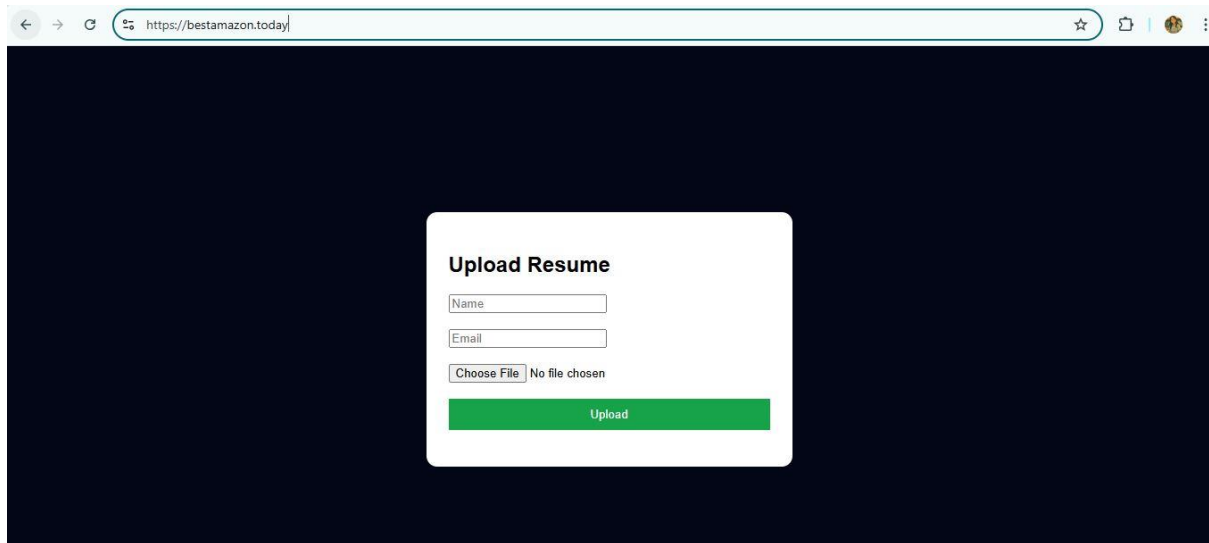
- HTTPS 443 → 0.0.0.0/0

STEP 19: Redirect HTTP → HTTPS

HTTP 80 → Redirect → HTTPS 443

FINAL TEST

Open browser: <https://bestamazon.today>



A screenshot of a web browser showing the 'Upload Resume' form on the website <https://bestamazon.today>. The form is centered on a dark blue background. It contains the following fields and buttons:

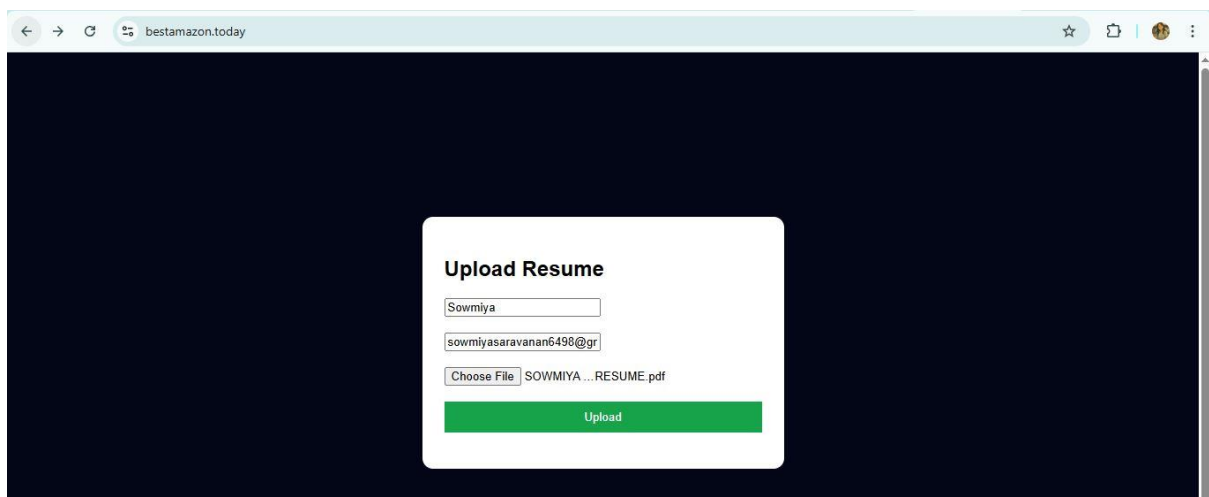
- Upload Resume** (Section Header)
-
-
- No file chosen
-

Final Output

- ✓ Web page loads successfully
- ✓ CV upload works
- ✓ File stored in S3
- ✓ Admin receives email
- ✓ HTTPS enabled
- ✓ Auto Scaling working

Screenshots of:

- **Upload page**



A screenshot of the same 'Upload Resume' form, but with the following data entered:

- Upload Resume** (Section Header)
-
-
- SOWMIYA ...RESUME.pdf
-

- **S3 object**

The screenshot shows the AWS Management Console interface for a bucket named 'resume-upload-demo-bucket'. The 'Objects' tab is active, displaying a list of objects. There is one object named 'Sowmiya_20251217071838.pdf' with a size of 262.3 KB and a storage class of 'Standard'. The object was last modified on December 17, 2025, at 12:48:39 (UTC+05:30). The interface includes various action buttons like 'Copy S3 URI', 'Copy URL', 'Download', 'Open', 'Delete', 'Actions', 'Create folder', and 'Upload'.

Name	Type	Last modified	Size	Storage class
Sowmiya_20251217071838.pdf	pdf	December 17, 2025, 12:48:39 (UTC+05:30)	262.3 KB	Standard

- **SNS email**

The screenshot shows a Gmail inbox with an email titled 'New Resume Uploaded' from 'resume-upload-topic'. The email was received 12:48 PM (0 minutes ago). The email content includes the name 'Sowmiya', the email address 'sowmiyasaravanan6498@gmail.com', and the file 'Sowmiya_20251217071838.pdf'. It also contains an unsubscribe link and a note to not reply directly to the email.

New Resume Uploaded inbox x

resume-upload-topic <no-reply@sns.amazonaws.com> to me

Name: Sowmiya
Email: sowmiyasaravanan6498@gmail.com
File: Sowmiya_20251217071838.pdf

...

If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:
<https://sns.ap-south-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:ap-south-1:353911359869:resume-upload-topic:65b63ccc-0b6d-4ac5-a2c6-272a027753da&Endpoint=sowmiyasaravanan6498@gmail.com>

Please do not reply directly to this email. If you have any questions or comments regarding this email, please contact us at <https://aws.amazon.com/support>